

# TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

## PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 12

**75 minutes**

### INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

**Calculators and dictionaries are NOT permitted.**

**There is no formulae booklet for this test.**

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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- 1** The function  $f(x) = 3\sin(2x) - 1$  is defined for all real  $x$ . Find the difference between its maximum and minimum values.

**A** 1

**B** 2

**C** 4

**D** 3

**E** 6

**F** 7

- 2 The trapezium rule with 2 strips of equal width is used to estimate  $\int_0^2 x^2 dx$ . Which of the following is true?
- A The estimate is  $\frac{8}{3}$  and it is exact.
  - B The estimate is 3 and it is an overestimate.
  - C The estimate is  $\frac{5}{2}$  and it is an overestimate.
  - D The estimate is  $\frac{5}{2}$  and it is an underestimate.
  - E The estimate is 3 and it is an underestimate.

- 3** The price of a coat is reduced by 20% in a sale, and then the sale price is reduced by a further 25%. The final price is what percentage of the original price?
- A** 50%
  - B** 55%
  - C** 65%
  - D** 60%
  - E** 40%
  - F** 45%

- 4** A chord of a circle of radius 13 is at perpendicular distance 5 from the centre. Find the length of the chord.
- A** 26
- B**  $\sqrt{144}$
- C** 18
- D** 12
- E** 10
- F** 24

5 Let  $f(x) = 2x + 1$  and  $g(x) = x^2 - 3$ . Find  $f(g(2)) - g(f(2))$ .

**A**     $-3$

**B**     $-17$

**C**     $-19$

**D**     $3$

**E**     $25$

**F**     $19$

**6** A car travels at 90 km/h. What is this speed in metres per second?

**A** 30 m/s

**B** 54 m/s

**C** 15 m/s

**D** 25 m/s

**E** 324 m/s

**F** 1.5 m/s

7 Given that  $\log_a x = 3$  and  $\log_a y = 5$ , find  $\log_a \left( \frac{x^2 \sqrt{y}}{a} \right)$ .

- A 8
- B  $\frac{15}{2}$
- C  $\frac{13}{2}$
- D  $\frac{17}{2}$
- E 10
- F  $\frac{11}{2}$

**8** If  $x = 2^p$ ,  $y = 2^q$  and  $\frac{x^3}{y^2} = 2^5$  with  $xy = 2^{10}$ , find  $p$ .

**A** 3

**B** 7

**C** 8

**D** 4

**E** 6

**F** 5

**9** Simplify  $\frac{1 - \cos^2 \theta}{\cos \theta \tan \theta}$  for values of  $\theta$  where the expression is defined.

**A** 1

**B**  $\sin \theta$

**C**  $\tan \theta$

**D**  $\sin^2 \theta$

**E**  $\cos \theta$

**F**  $\frac{\sin \theta}{\cos \theta}$

- 10** Which of the following could be the equation of a cubic curve that touches the  $x$ -axis at  $x = 2$  and crosses it at  $x = -1$ , and has a negative  $y$ -intercept?

**A**  $y = -(x - 2)^2(x + 1)$

**B**  $y = (x - 2)(x + 1)^2$

**C**  $y = (x + 2)^2(x - 1)$

**D**  $y = -(x + 2)^2(x - 1)$

**E**  $y = (x - 2)^2(x + 1)$

**11** The curve  $y = x^2 - 2ax + a^2 + a - 6$  has its vertex on the  $x$ -axis. Find the value of  $a$ .

**A** 0

**B** 6

**C** -6

**D** 3

**E** 2

**F** -2

- 12** In triangle  $ABC$ ,  $D$  lies on  $AB$  and  $E$  lies on  $AC$  with  $DE$  parallel to  $BC$ .  $AD = 4$ ,  $DB = 6$  and  $DE = 5$ . Find  $BC$ .

**A** 12.5

**B** 8

**C** 15

**D** 10

**E** 7.5

**F** 12

**13** Find the set of values of  $x$  satisfying both  $3x - 7 < 2$  and  $x^2 > 4$ .

**A**  $x < -2$  or  $2 < x < 3$

**B**  $2 < x < 3$

**C**  $-2 < x < 3$

**D**  $x < 3$

**E**  $x > 2$

**F**  $x < -2$

**14** A sequence satisfies  $x_{n+1} = \frac{1}{1-x_n}$  with  $x_1 = 2$ . Find  $x_{100}$ .

**A** 1

**B** 2

**C**  $\frac{1}{2}$

**D**  $-\frac{1}{2}$

**E** 0

**F** -1

- 15** The point  $P(x, y)$  moves so that its distance from  $(3, 0)$  is twice its distance from  $(0, 0)$ . The locus of  $P$  is a circle. Find its radius.

**A**  $\sqrt{2}$

**B** 3

**C** 2

**D** 4

**E**  $\sqrt{3}$

**F** 1

- 16** Simplify  $\frac{x^2 - 9}{(x+3)(x-4)} \times \frac{x^2 - 16}{x^2 - 3x}$ .
- A**  $\frac{x(x-3)}{x-4}$
- B**  $\frac{x-3}{x+3}$
- C**  $\frac{x}{x-4}$
- D**  $\frac{x}{(x+3)(x+4)}$
- E**  $\frac{x(x-3)}{(x-3)(x-4)}$
- F**  $\frac{x(x+3)}{x(x+3)}$

**17** Given  $f(x) = x^{\frac{3}{2}} - 6x^{\frac{1}{2}}$  for  $x > 0$ , find the value of  $x$  at which  $f$  has its minimum.

**A** 2

**B** 4

**C** 1

**D** 6

**E**  $\sqrt{2}$

**F** 3

**18** For how many integer values of  $k$  does  $x^2 + kx + 12 = 0$  have integer roots?

**A** 8

**B** 4

**C** 2

**D** 12

**E** 3

**F** 6

**19** The trapezium rule is used to estimate  $\int_a^b f(x) dx$ . For which of the following functions is the estimate guaranteed to be an underestimate for every choice of  $a < b$ ?

**A**  $f(x) = -x^2$

**B**  $f(x) = x^4$

**C**  $f(x) = 2x + 1$

**D**  $f(x) = x^2$

**E**  $f(x) = x^3$

**20** The coefficient of  $x^2$  in  $(1+x)^n + (1+x)^{n+1}$  is 36. Find  $n$ .

**A** 4

**B** 8

**C** 6

**D** 5

**E** 9

**F** 7

**END OF TEST**